Solving the Model Gap — Execution Summary

AMD-Sourcing the Engineering Cycles to Close NVIDIA's Physical-AI Lead

AMD hardware already wins on economics — roughly 50% lower capex and 30% lower opex than the incumbent, on competitive silicon (MI300X, Ryzen AI, Kria). Yet application developers still say no, because their products are already built around NVIDIA's models and pipelines (NGC, TensorRT, Isaac ROS, cuRobo, Isaac Sim, Cosmos). NVIDIA's moat is roughly 80% ecosystem and 20% hardware: 2M+ robotics developers, a 7+-year head start. Re-hosting a deployment costs 6–18 engineer-months and 5–7 re-targeted artifacts per model — a bill the customer, not AMD, has to pay before hardware economics even enter the conversation. That burden falls hardest on the small and mid-size robotics companies AMD most needs to win. Internally, this gap is known and winnable but cannot be closed by any single business unit: model training is iterative, the payoff is multi-year, and no BU can book near-term ROI on it — so, rationally, nobody starts, and the gap widens.

The way out is to pose this as a *replication problem, not an invention problem*. The target models are named, mostly open, and their performance is already published on NGC — we know exactly what "done" looks like. Only two things are missing: compute, which AMD already has (ALOLA and other AMD-wide clusters), and data, the real work (synthetic generation, pooled and curated open data, semi-supervised labeling, pseudo-labels from large open-source models and data pools). The proposal is an AMD-wide model training challenge: pick a target model, train it on AMD compute, submit a per-model leaderboard, and have PAVS validate and publish the result. This pattern has closed comparable capability gaps before — ImageNet seeded the deep-learning era, the DARPA Grand Challenge seeded self-driving, and Kaggle/Netflix Prize crowdsourced solutions that beat in-house teams at a fraction of the cost. A scored, public target repeatedly does what no single roadmap can.

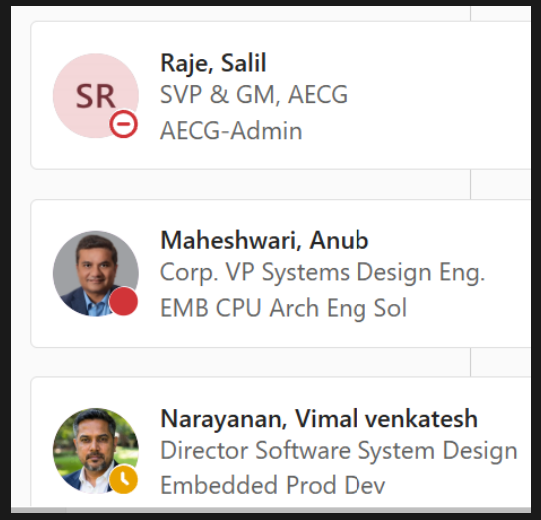
The mechanism is self-sustaining because every party wins by playing: participating engineers get recognition, a publishable track record, and a countable contribution to AMD; PAVS becomes the customer gateway for physical AI and the bridge across AMD's distributed teams; and AMD closes a strategic ecosystem gap with pooled compute and no single-BU budget line, building a pipeline of market-ready, AMD-native models and datasets along the way. The prize is worth the effort — AMD projects a $200B physical-AI silicon market by 2035, robotics startup funding hit $22.2B in 2025 (+69% YoY). Of 125 NGC models scanned, 48 across four priority tiers need an AMD-native replica (Appendix, Table A1–A4); Tier 1 + Tier 2 (30 models) is the actual "customer has this running on NVIDIA today and

The Ask

The plan is already staffed — we only need the mandate to start

What we need from leadership

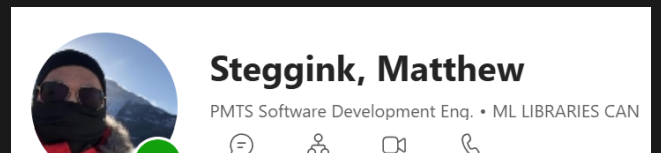
> An **org-wide email** endorsing the challenge — the signal that gives the effort **good traction**.



The push

Already staffed & committed — execution is ready

Compute — grants cluster access (ALOLA & AMD-wide)



AMD
together we advance_

Appendix A1 · Tier 1 — Production Perception Stack (NVIDIA Metropolis/DeepStream)

Highest priority — most commonly deployed NGC models a customer runs today and cannot carry over to AMD.

NVIDIA model	Function	Why it matters
TrafficCamNet	Vehicle detection (traffic cameras)	Core Metropolis/smart-city SKU
PeopleNet / PeopleNet-AMR	Person detection	Most widely deployed DeepStream model
License Plate Recognition (LPRNet) + LPDNet	Plate detection + recognition pair	Sold as a pair, tolling/parking/security
VehicleMakeNet / VehicleTypeNet	Vehicle attribute classification	Paired with TrafficCamNet in the same pipelines
ReIdentificationNet (+ Transformer variant)	Person re-identification across cameras	Retail/security analytics
PeopleSemSegNet / PeopleSemSegNet AMR	Person segmentation	Privacy-masking, robotics safety zones
Retail Object Detection / Retail Object Recognition	Shelf/product detection	Retail analytics vertical
Action Recognition Net	Action/behavior classification	Security, workplace safety
BodyPoseNet / PoseClassificationNet	Human pose estimation	Robotics, ergonomics, fitness
Optical Character Detection/Recognition (OCDNet/OCRNet)	Scene text OCR	Logistics, document/label reading

Appendix A2 · Tier 2 — Robotics/Perception Building Blocks (Physical AI)

The models Isaac ROS perception pipelines are built on — closest to AMD-ROS/AARTS.

NVIDIA model	Function
Bi3D Proximity Segmentation	Stereo-based obstacle proximity
ESS DNN Stereo Disparity	Stereo depth
FoundationPose	6-DoF object pose estimation/tracking
PointPillarNet	LiDAR 3D object detection
Transfusion for 3D Object Detection	Camera+LiDAR fusion detection
CitySemSegFormer	Semantic segmentation (autonomy/city-scale)
TAO Pretrained Classification/DetectNet V2/EfficientDet/Instance Seg/Semantic Seg	Generic backbones customers fine-tune on top of — AMD needs *some* pretrained backbone family here or customers can't even start a TAO-style workflow on AMD

> Tier 1 + Tier 2 = **30 models** — the real "customer has this running on NVIDIA today and can't move it" set.

Appendix A3 · Tier 3 — Biometric / Human-Sensing (Smaller but Sticky Niche)

Matters mainly if AMD is also chasing the biometric/kiosk vertical.

NVIDIA model	Function
FaceNet	Face detection
FaceDetectIR	Face detection (infrared)
GazeNet (Gaze Estimation)	Gaze tracking
EmotionNet	Emotion classification
GestureNet	Hand/body gesture recognition
HeartRateNet	Remote heart-rate estimation
Facial Landmark Estimator (FPENet)	Facial landmark detection
Pre-trained Deformable DETR NvImageNet Weights	Detection backbone
Pre-trained FAN based NVImageNet Classification Weights	Classification backbone
Pre-trained GCViT NVImageNet Classification Weights	Classification backbone
Visual ChangeNet Classification	Change detection classification
NVIDIA Optical Inspection Model	Industrial visual inspection

Appendix A4 · Tier 4 — Enterprise Vision & HPC/Climate (Lowest Priority)

Matters mainly if AMD is also chasing the enterprise-vision or HPC-climate vertical.

NVIDIA model	Function	Why lower priority
PCB Defect Classification Model	Industrial inspection	NVIDIA AI Enterprise EULA; small footprint
PeopleSemSegFormer Model Card	Person segmentation	NVIDIA AI Enterprise EULA; small footprint
Pre-trained DINO NvImageNet Weights	Classification backbone	NVIDIA AI Enterprise EULA; small footprint
PhysicsNeMo Checkpoints: AFNO_DX_FI-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance
PhysicsNeMo Checkpoints: AFNO_DX_SR-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance
PhysicsNeMo Checkpoints: AFNO_DX_TP-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance
PhysicsNeMo Checkpoints: AFNO_DX_WG-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance
PhysicsNeMo Checkpoints: DLESyM-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance

48 models across 4 tiers, out of 125 scanned on NGC — the other 77 are either already portable (49) or unresolved catalog entries.